UML Diagrams

g-code-viewer

Black Sphere Industries

Generated 2026-03-23 | D2 Diagrams

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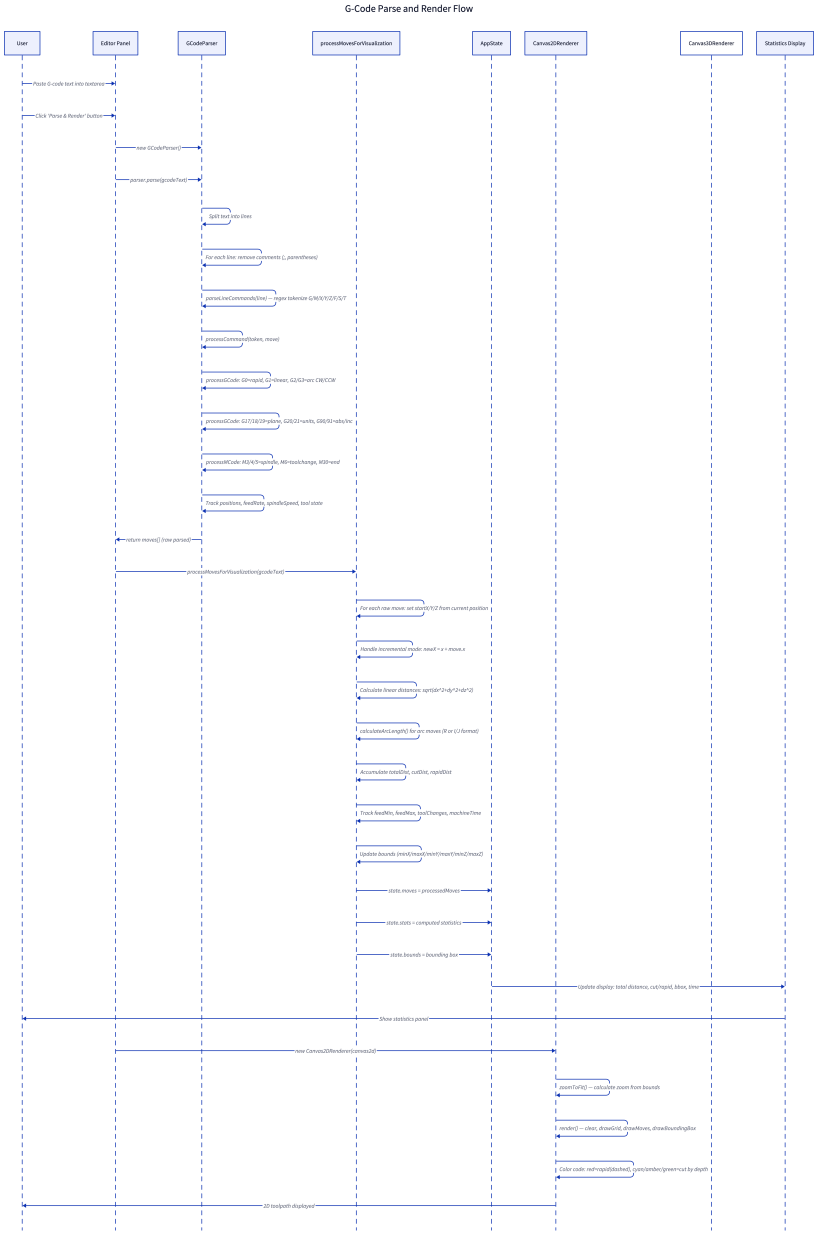
```
class-diagram: {
  GCodeParser: GCodeParser {
    shape: class
    units: string
    feedMode: string
    plane: string
    spindle: boolean
    spindleDir: number
    positions: {x, y, z, a, b, c}
    offsets: {x, y, z}
    feedRate: number
    spindleSpeed: number
    tool: number
    moves: "Move[]"
    parse(gcodeText)
    parseLineCommands(line)
    processCommand(cmd, move)
    processGCode(value, move)
    processMCode(value, move)
  }

  AppState: Global State {
    shape: class
    gcode: "Move[]"
    moves: "ProcessedMove[]"
    currentLine: number
    isPlaying: boolean
    isPaused: boolean
    speed: number
    toolDiameter: number
    units: string
    currentX: number
    currentY: number
    currentZ: number
    currentF: number
    currentS: number
    currentT: number
    feedMode: string
    plane: string
    bounds: BoundingBox
    stats: Statistics
  }

  Canvas2DRenderer: Canvas2DRenderer {
    shape: class
    canvas: HTMLCanvasElement
    ctx: CanvasRenderingContext2D
    panX: number
    panY: number
    zoom: number
    dpr: number
    setup()
    render()
    drawGrid(ctx)
    drawMoves(ctx)
    drawArc(ctx, move)
    drawBoundingBox(ctx)
```

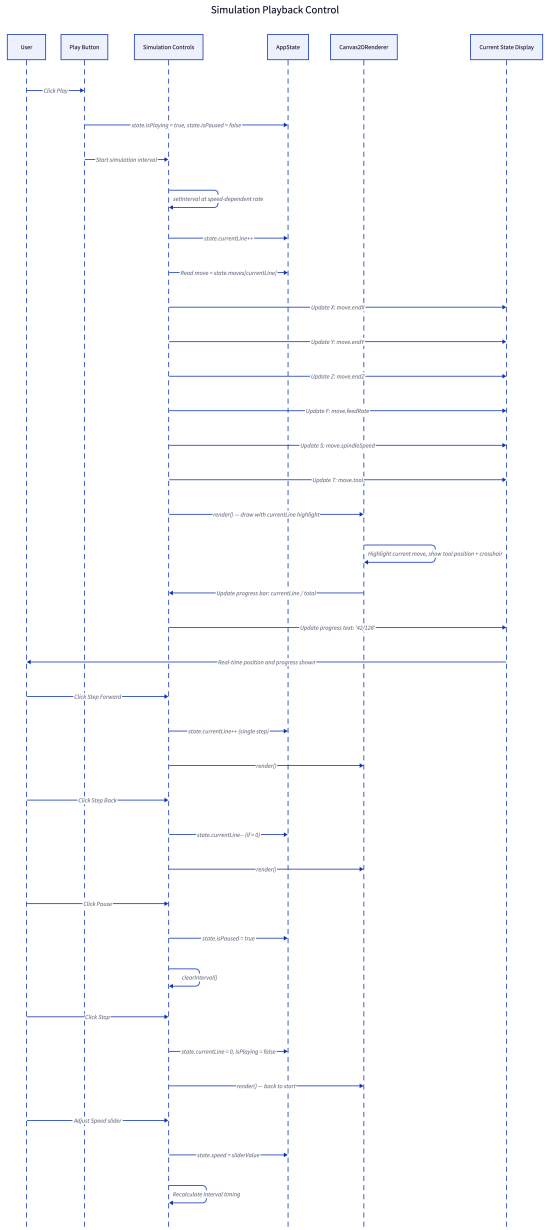
```
    handleZoom(event)
    handlePan(event)
    zoomToFit()
  }

  Canvas3DRenderer: Canvas3DRenderer {
    shape: class
    canvas: HTMLCanvasElement
    ctx: CanvasRenderingContext2D
    rotX: number
    rotZ: number
    zoom: number
    panX: number
    panY: number
    setup()
    render()
    project(x, y, z)
    subdivideArc(move)
    handleZoom(event)
    handleRotate(event)
    zoomToFit()
  }
```

... (148 lines total)

G-Code Parse and Render Flow



Simulation Playback Control



File Upload and Example Loading

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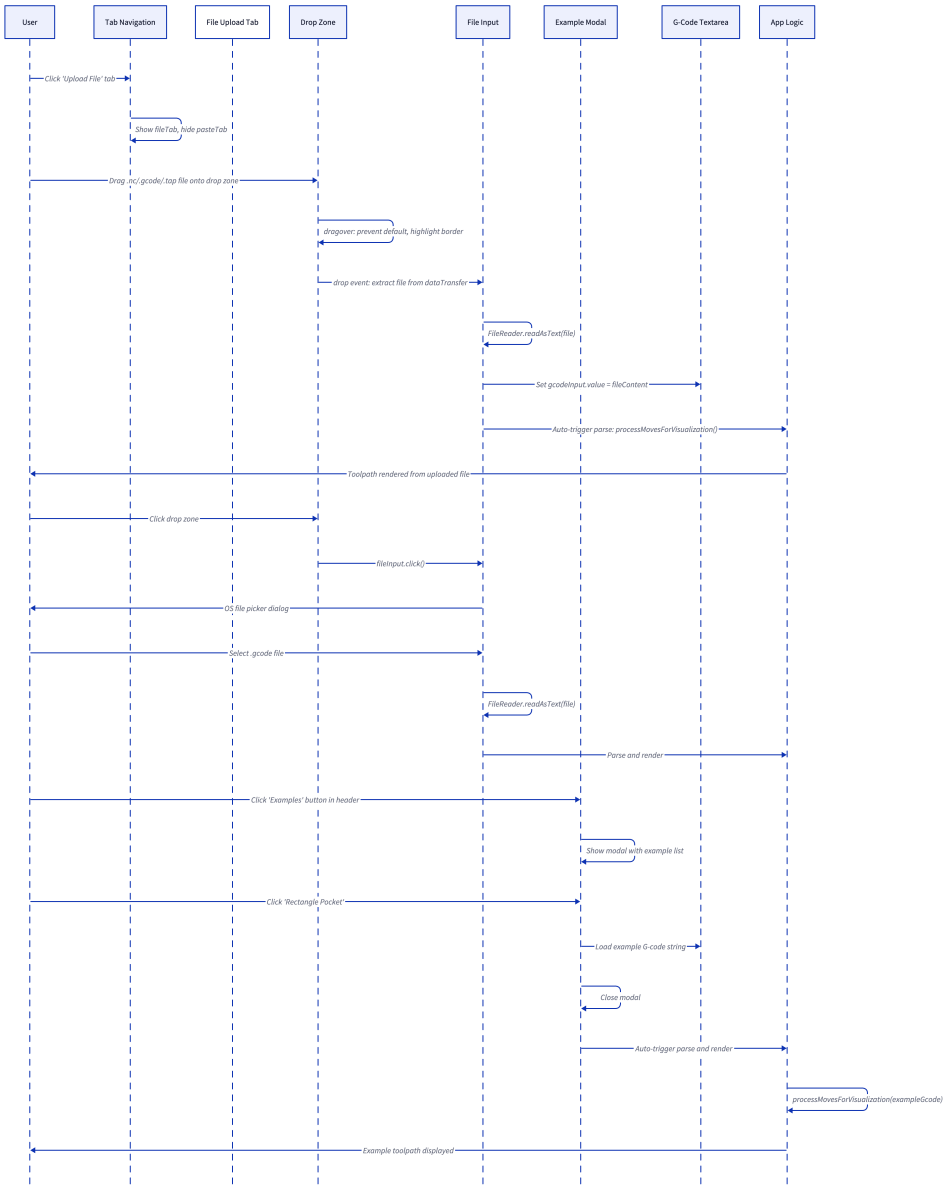
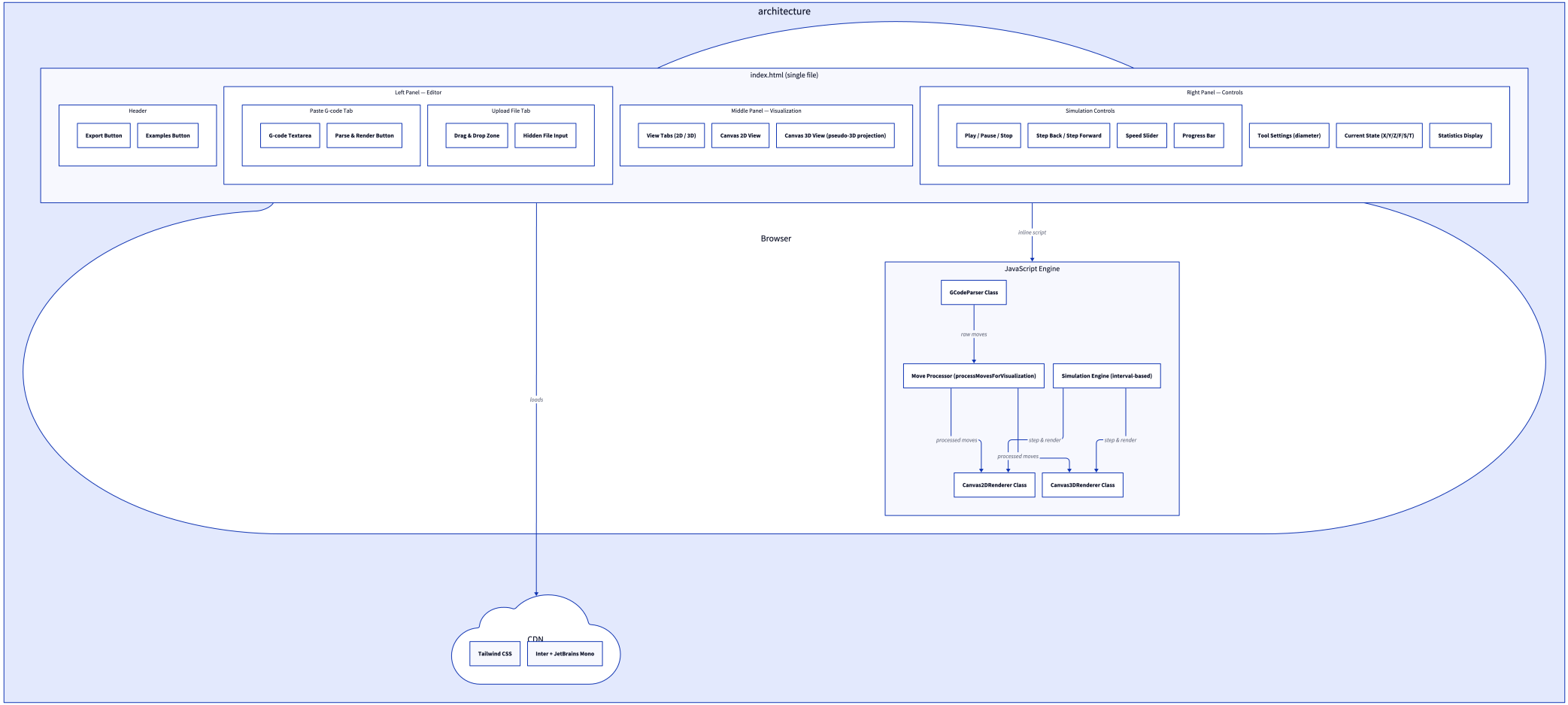
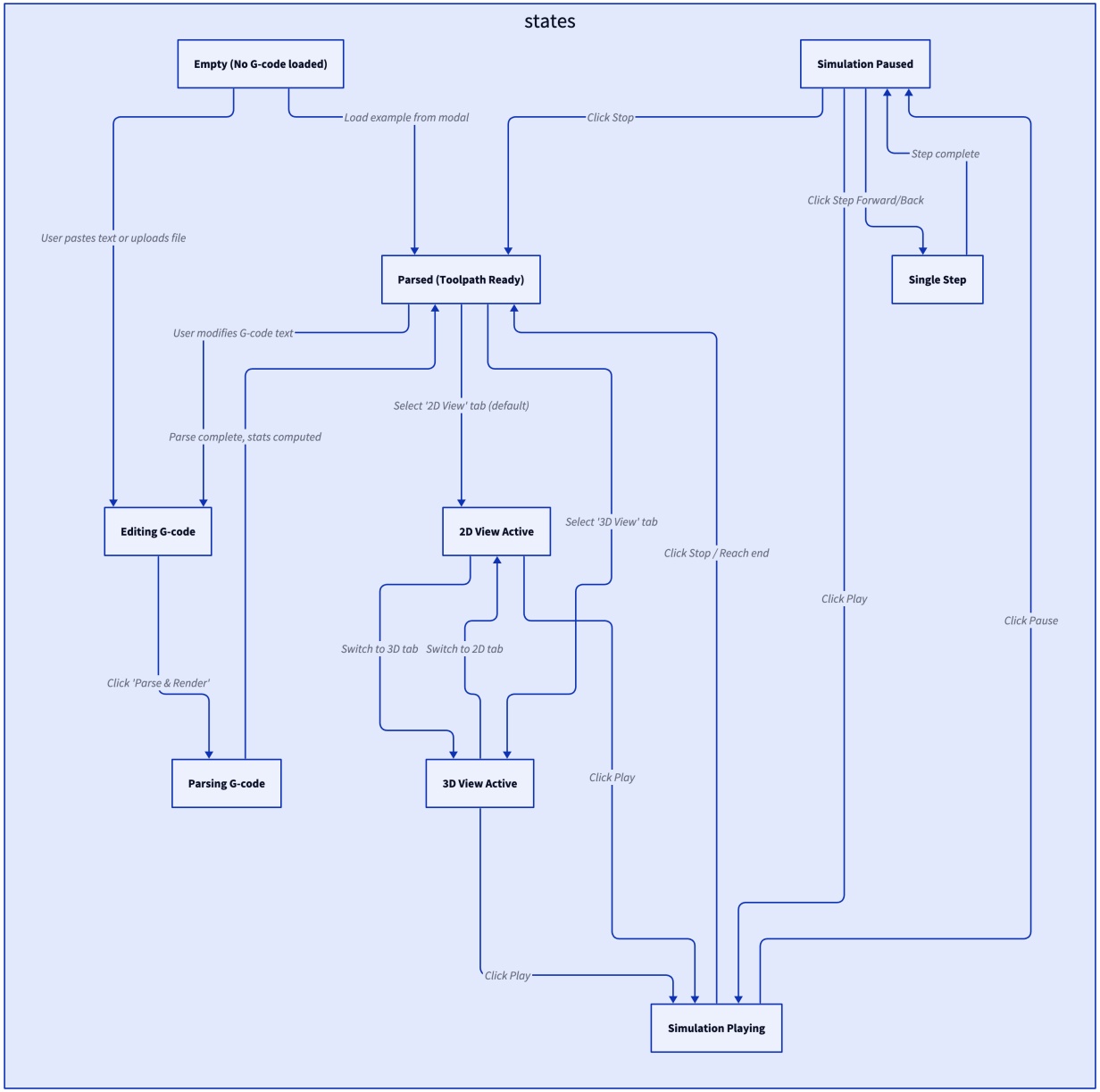



architecture
index.html (single file)
Header
Export Button
Examples Button
Left Panel — Editor
Paste G-code Tab
G-code Textarea
Parse & Render Button
Upload File Tab
Drag & Drop Zone
Hidden File Input
Middle Panel — Visualization
View Tabs (2D / 3D)
Canvas 2D View
Canvas 3D View (pseudo-3D projection)
Right Panel — Controls
Simulation Controls
Play / Pause / Stop
Step Back / Step Forward
Speed Slider
Progress Bar
Tool Settings (diameter)
Current State (X/Y/Z/F/S/T)
Statistics Display
Browser
inline script
JavaScript Engine
GCodeParser Class
raw moves
Move Processor (processMovesForVisualization)
Simulation Engine (interval-based)
processed moves
step & render
step & render
processed moves
Canvas2DRenderer Class
Canvas3DRenderer Class
loads
CDN
Tailwind CSS
Inter + JetBrains Mono


states
Empty (No G-code loaded)
Simulation Paused
Load example from modal
Click Stop
Step complete
Click Step Forward/Back
User pastes text or uploads file
Parsed (Toolpath Ready)
Single Step
User modifies G-code text
Select '2D View' tab (default)
Parse complete, stats computed
Editing G-code
2D View Active
Select '3D View' tab
Click Stop / Reach end
Click Play
Click Pause
Switch to 3D tab
Switch to 2D tab
Click 'Parse & Render'
Parsing G-code
3D View Active
Click Play
Click Play
Simulation Playing